

Multiple Choice (10 marks)

1. A subset H of a group $(G, *)$ is a group if
 - (a) $a, b \in H \Rightarrow a * b \in H$
 - (b) $a \in H \Rightarrow a^{-1} \in H$
 - (b) $a, b \in H \Rightarrow a * b^{-1} \in H$
 - (b) H contains the identity element
2. In the xy -plane, the curve with parametric equations $x = \cos t$ and $y = \sin t$, $0 \leq t \leq \pi$, has length
 - (a) 3
 - (b) π
 - (c) 3π
 - (d) $3/2$
3. If the finite group G contains a subgroup of order seven but no element (other than the identity) is its own inverse, then the order of G could be
 - (a) 27
 - (b) 28
 - (c) 35
 - (d) 37
4. Statement 1 | $\mathbb{R}$ is a splitting field of some polynomial over $\mathbb{Q}$. Statement 2 | There is a field with 60 elements.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
5. Find the sum of the given polynomials in the given polynomial ring. $f(x) = 4x - 5$, $g(x) = 2x^2 - 4x + 2$ in $\mathbb{Z}_8[x]$.
 - (a) $2x^2 + 5$
 - (b) $6x^2 + 4x + 6$
 - (c) 0
 - (d) $x^2 + 1$

True / False (10 marks)

Answer True or False

6. True or False: The maximum possible order for an element of S_{10} is 30.
7. True or False: The degree of the field extension $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbb{Q}$ is 2.
8. True or False: The linear transformation T maps the point $(2, 1)$ to the point $(-1, 4)$.
9. True or False: The index of the subgroup generated by the permutation p in S_5 is 8.
10. True or False: The identity element of the group formed by the integers $\mathbb{Z}$ under the operation $a*b = a + b + 1$ is 0.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the properties of a linear transformation P on a finite-dimensional real vector space V given that $P^2 = P$. Analyze whether P is always diagonalizable and explore the implications of this property.
12. Given the circle C in the yz -plane defined by the equation $(y - 3)^2 + z^2 = 1$, derive the equation of the torus formed when C is revolved around the z -axis, and explain the geometric implications of this transformation.

End of Paper